Simultaneous constructive order-code relations for higher inductive-inductive surreal numbers

The Univalent Foundations Program

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Editorial note. The fixed official author source and CC BY-SA 3.0 Unported attribution are preserved. This task migrates S046-defn:No-codes once.

Editorial note. The original proof ends at2909 and explicitly treats the last left-transitivity cases by symmetry; the entire author proof is retained. Its reference to the Cauchy approximation proof is comparative and supplies no missing lemma.

Editorial note. 2026-10-11 材料收口：复用既有 H2、来源许可和近邻判断；核对原命题与作者证明起止、所指上下文及装配范围齐备。未重证作者结论；沿用既有非重复依据。

Allowed standard prerequisites (editorial).

- Univalent type theory and supplied HIIT elimination. Dependent sums and products; mere propositions and their truncation elimination; function extensionality and univalence for propositions; the supplied higher inductive-inductive surreal constructors and mutual elimination rules. A reflexive mere relation implying identity makes its type a set.

Source reference. Original theorem 11.3.16: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1177–1177. Comparison of proof organization only; the surreal proof constructs its own motives and completes all of its own cases. Cauchy approximation is not used as a theorem.

Source reference. Original theorem 7.2.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1797–1797. Explicitly allowed basic sethood criterion: a

reflexive mere relation implying identity makes a type a set; not an order-code construction.

Definition 11.6.1. The type No of **surreal numbers**, along with the relations $< : \mathsf{No} \to \mathsf{No} \to \mathcal{U}$ and $\leq : \mathsf{No} \to \mathsf{No} \to \mathcal{U}$, are defined higher inductive-inductively as follows. The type No has the following constructors.

- For any $\mathcal{L}, \mathcal{R} : \mathcal{U}$ and functions $\mathcal{L} \to \mathsf{No}$ and $\mathcal{R} \to \mathsf{No}$, whose values we write as x^L and x^R for $L : \mathcal{L}$ and $R : \mathcal{R}$ respectively, if $\forall(L : \mathcal{L}).\forall(R : \mathcal{R}).\, x^L < x^R$, then there is a surreal number x.
- For any $x, y : \mathsf{No}$ such that $x \leq y$ and $y \leq x$, we have $\mathsf{eq}_{\mathsf{No}}(x, y) : x = y$.

We will refer to the inputs of the first constructor as a **cut**. If x is the surreal number constructed from a cut, then the notation x^L will implicitly assume $L : \mathcal{L}$, and similarly x^R will assume $R : \mathcal{R}$. In this way we can usually avoid naming the indexing types $\mathcal{L}$ and $\mathcal{R}$, which is convenient when there are many different cuts under discussion. Following Conway, we call x^L a *left option* of x and x^R a *right option*.

The path constructor implies that different cuts can define the same surreal number. Thus, it does not make sense to speak of the left or right options of an arbitrary surreal number x, unless we also know that x is defined by a particular cut. Thus in what follows we will say, for instance, "given a cut defining a surreal number x" in contrast to "given a surreal number x".

The relation $\leq$ has the following constructors.

- Given cuts defining two surreal numbers x and y, if $x^L < y$ for all L, and $x < y^R$ for all R, then $x \leq y$.
- Propositional truncation: for any $x, y : \mathsf{No}$, if $p, q : x \leq y$, then $p = q$.

And the relation $<$ has the following constructors.

- Given cuts defining two surreal numbers x and y, if there is an L such that $x \leq y^L$, then $x < y$.
- Given cuts defining two surreal numbers x and y, if there is an R such that $x^R \leq y$, then $x < y$.
- Propositional truncation: for any $x, y : \mathsf{No}$, if $p, q : x < y$, then $p = q$.

In order to say much more about surreal numbers, however, we need their induction principle. The mutual induction principle for $(\mathsf{No}, \leq, <)$ applies to

three families of types:

$$\begin{aligned} A &: \mathsf{No} \to \mathcal{U} \\ B &: \prod_{(x,y:\mathsf{No})} \prod_{(a:A(x))} \prod_{(b:A(y))} (x \leq y) \to \mathcal{U} \\ C &: \prod_{(x,y:\mathsf{No})} \prod_{(a:A(x))} \prod_{(b:A(y))} (x < y) \to \mathcal{U}. \end{aligned}$$

As with the induction principle for Cauchy reals, it is helpful to think of B and C as families of relations between the types $A(x)$ and $A(y)$. Thus we write $B(x, y, a, b, \xi)$ as $(x, a) \preccurlyeq^{\xi} (y, b)$ and $C(x, y, a, b, \xi)$ as $(x, a) \prec^{\xi} (y, b)$. Similarly, we usually omit the ξ since it inhabits a mere proposition and so is uninteresting, and we may often omit x and y as well, writing simply $a \preccurlyeq b$ or $a \prec b$. With these notations, the hypotheses of the induction principle are the following.

- For any cut defining a surreal number x, together with

  (i) for each L, an element $a^L : A(x^L)$, and
  (ii) for each R, an element $a^R : A(x^R)$, such that
  (iii) for all L and R we have $(x^L, a^L) \prec (x^R, a^R)$

  there is a specified element $f_a : A(x)$. We call such data a **dependent cut** over the cut defining x.
- For any $x, y : \mathsf{No}$ with $a : A(x)$ and $b : A(y)$, if $x \leq y$ and $y \leq x$ and also $(x, a) \preccurlyeq (y, b)$ and $(y, b) \preccurlyeq (x, a)$, then $a =^{A}_{\mathsf{eq_{No}}} b$.
- Given cuts defining two surreal numbers x and y, and dependent cuts a over x and b over y, such that for all L we have $x^L < y$ and $(x^L, a^L) \prec (y, f_b)$, and for all R we have $x < y^R$ and $(x, f_a) \prec (y^R, b^R)$, then $(x, f_a) \preccurlyeq (y, f_b)$.
- $\preccurlyeq$ takes values in mere propositions.
- Given cuts defining two surreal numbers x and y, dependent cuts a over x and b over y, and an L_0 such that $x \leq y^{L_0}$ and $(x, f_a) \preccurlyeq (y^{L_0}, b^{L_0})$, we have $(x, f_a) \prec (y, f_b)$.
- Given cuts defining two surreal numbers x and y, dependent cuts a over x and b over y, and an R_0 such that $x^{R_0} \leq y$ together with $(x^{R_0}, a^{R_0}), \preccurlyeq (y, f_b)$, we have $(x, f_a) \prec (y, f_b)$.
- $\prec$ takes values in mere propositions.

Under these hypotheses we deduce a function $f : \prod_{(x:\mathsf{No})} A(x)$ such that

$$\begin{aligned} f(x) &\equiv f_{f[x]} \\ (x \le y) &\Rightarrow (x, f(x)) \trianglelefteq (y, f(y)) \\ (x < y) &\Rightarrow (x, f(x)) \triangleleft (y, f(y)). \end{aligned} \tag{11.6.2}$$

In the computation rule (11.6.2) for the point constructor, x is a surreal number defined by a cut, and $f[x]$ denotes the dependent cut over x defined by applying f (and using the fact that f takes $<$ to $\triangleleft$). As usual, we will generally use pattern-matching notation, where the definition of f on a cut $\{\, x^L \mid x^R \,\}$ may use the symbols $f(x^L)$ and $f(x^R)$ and the assumption that they form a dependent cut.

As with the Cauchy reals, we have special cases resulting from trivializing some of A, $\trianglelefteq$, and $\triangleleft$. Taking $\trianglelefteq$ and $\triangleleft$ to be constant at $\mathbf{1}$, we have **No-induction**, which for simplicity we state only for mere properties:

- Given $P : \mathsf{No} \to \mathsf{Prop}$, if $P(x)$ holds whenever x is a surreal number defined by a cut such that $P(x^L)$ and $P(x^R)$ hold for all L and R, then $P(x)$ holds for all $x : \mathsf{No}$.

Theorem 11.6.3 (Conway's Theorem 0)**.**

(i) For any $x : \mathsf{No}$, we have $x \le x$.

(ii) For any $x : \mathsf{No}$ defined by a cut, we have $x^L < x$ and $x < x^R$ for all L and R.

证明. Note first that if $x \le x$, then whenever x occurs as a left option of some cut y, we have $x < y$ by the first constructor of $<$, and similarly whenever x occurs as a right option of a cut y, we have $y < x$ by the second constructor of $<$. In particular, (i)⇒(ii).

We prove (i) by No-induction on x. Thus, assume x is defined by a cut such that $x^L \le x^L$ and $x^R \le x^R$ for all L and R. But by our observation above, these assumptions imply $x^L < x$ and $x < x^R$ for all L and R, yielding $x \le x$ by the constructor of $\le$. □

Corollary 11.6.4. No *is a 0-type.*

证明. The mere relation $R(x, y) :\equiv (x \le y) \wedge (y \le x)$ implies identity by the path constructor of No, and contains the diagonal by Theorem 11.6.3(i). Thus, original theorem 7.2.2 applies. □

We will also need the joint recursion principle, $(\mathsf{No}, \le, <)$**-recursion**. It is convenient to state this as follows. Suppose A is a type equipped with relations

$\preccurlyeq : A \to A \to \mathsf{Prop}$ and $\lhd : A \to A \to \mathsf{Prop}$. Then we can define $f : \mathsf{No} \to A$ by doing the following.

(i) For any x defined by a cut, assuming $f(x^L)$ and $f(x^R)$ to be defined such that $f(x^L) \lhd f(x^R)$ for all L and R, we must define $f(x)$. (We call this the *primary clause* of the recursion.)

(ii) Prove that $\preccurlyeq$ is *antisymmetric*: if $a \preccurlyeq b$ and $b \preccurlyeq a$, then $a = b$.

(iii) For x, y defined by cuts such that $x^L < y$ for all L and $x < y^R$ for all R, and assuming inductively that $f(x^L) \lhd f(y)$ for all L, $f(x) \lhd f(y^R)$ for all R, and also that $f(x^L) \lhd f(x^R)$ and $f(y^L) \lhd f(y^R)$ for all L and R, we must prove $f(x) \preccurlyeq f(y)$.

(iv) For x, y defined by cuts and an L_0 such that $x \leq y^{L_0}$, and assuming inductively that $f(x) \preccurlyeq f(y^{L_0})$, and also that $f(x^L) \lhd f(x^R)$ and $f(y^L) \lhd f(y^R)$ for all L and R, we must prove $f(x) \lhd f(y)$.

(v) For x, y defined by cuts and an R_0 such that $x^{R_0} \leq y$, and assuming inductively that $f(x^{R_0}) \preccurlyeq f(y)$, and also that $f(x^L) \lhd f(x^R)$ and $f(y^L) \lhd f(y^R)$ for all L and R, we must prove $f(x) \lhd f(y)$.

The last three clauses can be more concisely described by saying we must prove that f (as defined in the first clause) takes $\leq$ to $\preccurlyeq$ and $<$ to $\lhd$. We will refer to these properties by saying that f *preserves inequalities*. Moreover, in proving that f preserves inequalities, we may assume the particular instance of $\leq$ or $<$ to be obtained from one of its constructors, and we may also use inductive hypotheses that f preserves all inequalities appearing in the input to that constructor.

If we succeed at (i)–(v) above, then we obtain $f : \mathsf{No} \to A$, which computes on cuts as specified by (i), and which preserves all inequalities:

$$\forall(x, y : \mathsf{No}).\, \Big((x \leq y) \to (f(x) \preccurlyeq f(y))\Big) \wedge \Big((x < y) \to (f(x) \lhd f(y))\Big).$$

Like $(\mathbb{R}_{\mathsf{c}}, \sim)$-recursion for the Cauchy reals, this recursion principle is essential for defining functions on No, since we cannot first define a function on "pre-surreals" and only later prove that it respects the notion of equality.

Theorem 11.6.5. *There are relations* $\preceq : \mathsf{No} \to \mathsf{No} \to \mathsf{Prop}$ *and* $\prec : \mathsf{No} \to \mathsf{No} \to \mathsf{Prop}$ *such that if* x *and* y *are surreals defined by cuts, then*

$$(x \preceq y) :\equiv \big(\forall(L).\, x^L \prec y\big) \wedge \big(\forall(R).\, x \prec y^R\big)$$
$$(x \prec y) :\equiv \big(\exists(L).\, x \preceq y^L\big) \vee \big(\exists(R).\, x^R \preceq y\big).$$

Moreover, we have

$$(x \prec y) \to (x \preceq y) \tag{11.6.6}$$

and all the reasonable transitivity properties making $\prec$ and $\preceq$ into a "bimodule" over $\leq$ and $<$:

$$\begin{aligned}
&(x \leq y) \to (y \preceq z) \to (x \preceq z) && (x \preceq y) \to (y \leq z) \to (x \preceq z)\\
&(x \leq y) \to (y \prec z) \to (x \prec z) && (x \preceq y) \to (y < z) \to (x \prec z)\\
&(x < y) \to (y \preceq z) \to (x \prec z) && (x \prec y) \to (y \leq z) \to (x \prec z).
\end{aligned} \tag{11.6.7}$$

证明. We define $\preceq$ and $\prec$ by double $(\mathsf{No}, \leq, <)$-induction on x, y. The first induction is a simple recursion, whose codomain is the subset A of $(\mathsf{No} \to \mathsf{Prop}) \times (\mathsf{No} \to \mathsf{Prop})$ consisting of pairs of predicates of which one implies the other and which satisfy "transitivity on the right", i.e. (11.6.6) and the right column of (11.6.7) with $(x \preceq -)$ and $(x \prec -)$ replaced by the two given predicates. As in the proof of original theorem 11.3.16, we regard these predicates as half of binary relations, writing them as $y \mapsto (\diamondsuit \preceq y)$ and $y \mapsto (\diamondsuit \prec y)$, with $\diamondsuit$ denoting the pair of relations. We equip A with the following two relations:

$$\begin{aligned}
(\diamondsuit \trianglelefteq \heartsuit) &:\equiv \forall(y : \mathsf{No}).\, \Big((\heartsuit \preceq y) \to (\diamondsuit \preceq y)\Big) \wedge \Big((\heartsuit \prec y) \to (\diamondsuit \prec y)\Big),\\
(\diamondsuit \triangleleft \heartsuit) &:\equiv \forall(y : \mathsf{No}).\, \Big((\heartsuit \preceq y) \to (\diamondsuit \prec y)\Big).
\end{aligned}$$

Note that $\trianglelefteq$ is antisymmetric, since if $\diamondsuit \trianglelefteq \heartsuit$ and $\heartsuit \trianglelefteq \diamondsuit$, then $(\heartsuit \preceq y) \Leftrightarrow (\diamondsuit \preceq y)$ and $(\heartsuit \prec y) \Leftrightarrow (\diamondsuit \prec y)$ for all y, hence $\diamondsuit = \heartsuit$ by univalence for mere propositions and function extensionality. Moreover, to say that a function $\mathsf{No} \to A$ preserves inequalities is exactly to say that, when regarded as a pair of binary relations on No, it satisfies "transitivity on the left" (the left column of (11.6.7)).

Now for the primary clause of the recursion, we assume given x defined by a cut, and relations $(x^L \prec -)$, $(x^R \prec -)$, $(x^L \preceq -)$, and $(x^R \preceq -)$ for all L and R, of which the strict ones imply the non-strict ones, which satisfy transitivity on the right, and such that

$$\forall(L, R).\, \forall(y : \mathsf{No}).\, \Big((x^R \preceq y) \to (x^L \prec y)\Big). \tag{11.6.8}$$

We now have to define $(x \prec y)$ and $(x \preceq y)$ for all y. Here in contrast to original theorem 11.3.16, rather than a nested recursion, we use a nested induction, in order to be able to inductively use transitivity on the left with respect to the inequalities $x^L < x$ and $x < x^R$. Define $A' : \mathsf{No} \to \mathcal{U}$ by taking $A'(y)$ to be the subset A' of $\mathsf{Prop} \times \mathsf{Prop}$ consisting of two mere propositions, denoted $\triangle \preceq y$ and $\triangle \prec y$ (with $\triangle : A'(y)$), such that

$$(\triangle \prec y) \to (\triangle \preceq y) \tag{11.6.9}$$

$$\forall(L).\, (\triangle \preceq y) \to (x^L \prec y) \tag{11.6.10}$$

$$\forall(R).\, (x^R \preceq y) \to (\triangle \prec y). \tag{11.6.11}$$

Using notation analogous to $\trianglelefteq$ and $\triangleleft$, we equip A' with the two relations defined for $\triangle : A'(y)$ and $\square : A'(z)$ by

$$(\triangle \sqsubseteq \square) :\equiv \Big((\triangle \preceq y) \to (\square \preceq z)\Big) \wedge \Big((\triangle \prec y) \to (\square \prec z)\Big)$$
$$(\triangle \sqsubset \square) :\equiv \Big((\triangle \preceq y) \to (\square \prec z)\Big).$$

Again, $\sqsubseteq$ is evidently antisymmetric in the appropriate sense. Moreover, a function $\prod_{(y:\mathsf{No})} A'(y)$ which preserves inequalities is precisely a pair of predicates of which one implies the other, which satisfy transitivity on the right, and transitivity on the left with respect to the inequalities $x^L < x$ and $x < x^R$. Thus, this inner induction will provide what we need to complete the primary clause of the outer recursion.

For the primary clause of the inner induction, we assume also given y defined by a cut, and properties $(x \prec y^L)$, $(x \prec y^R)$, $(x \preceq y^L)$, and $(x \preceq y^R)$ for all L and R, with the strict ones implying the non-strict ones, transitivity on the left with respect to $x^L < x$ and $x < x^R$, and on the right with respect to $y^L < y^R$. We can now give the definitions specified in the theorem statement:

$$(x \preceq y) :\equiv (\forall(L).\, x^L \prec y) \wedge (\forall(R).\, x \prec y^R), \tag{11.6.12}$$
$$(x \prec y) :\equiv (\exists(L).\, x \preceq y^L) \vee (\exists(R).\, x^R \preceq y). \tag{11.6.13}$$

For this to define an element of $A'(y)$, we must show first that $(x \prec y) \to (x \preceq y)$. The assumption $x \prec y$ has two cases. On one hand, if there is L_0 with $x \preceq y^{L_0}$, then by transitivity on the right with respect to $y^{L_0} < y^R$, we have $x \prec y^R$ for all R. Moreover, by transitivity on the left with respect to $x^L < x$, we have $x^L \prec y^{L_0}$ for any L, hence $x^L \prec y$ by transitivity on the right. Thus, $x \preceq y$.

On the other hand, if there is R_0 with $x^{R_0} \preceq y$, then by (11.6.8), we have $x^L \prec y$ for all L. And by transitivity on the left and right with respect to $x < x^{R_0}$ and $y < y^R$, we have $x \prec y^R$ for any R. Thus, $x \preceq y$.

We also need to show that these definitions are transitive on the left with respect to $x^L < x$ and $x < x^R$. But if $x \preceq y$, then $x^L \prec y$ for all L by definition; while if $x^R \preceq y$, then $x \prec y$ also by definition.

Thus, (11.6.12) and (11.6.13) do define an element of $A'(y)$. We now have to verify that this definition preserves inequalities, as a dependent function into A', i.e. that these relations are transitive on the right. Remember that in each case, we may assume inductively that they are transitive on the right with respect to all inequalities arising in the inequality constructor.

- Suppose $x \preceq y$ and $y \leq z$, the latter arising from $y^L < z$ and $y < z^R$ for all L and R. Then the inductive hypothesis (of the inner recursion)

applied to $y < z^R$ yields $x \prec z^R$ for any R. Moreover, by definition $x \preceq y$ implies that $x^L \prec y$ for any L, so by the inductive hypothesis of the outer recursion we have $x^L \prec z$. Thus, $x \preceq z$.

- Suppose $x \preceq y$ and $y < z$. First, suppose $y < z$ arises from $y \leq z^{L_0}$. Then the inner inductive hypothesis applied to $y \leq z^{L_0}$ yields $x \preceq z^{L_0}$, hence $x \prec z$.

  Second, suppose $y < z$ arises from $y^{R_0} \leq z$. Then by definition, $x \preceq y$ implies $x \prec y^{R_0}$, and then the inner inductive hypothesis for $y^{R_0} \leq z$ yields $x \prec z$.
- Suppose $x \prec y$ and $y \leq z$, the latter arising from $y^L < z$ and $y < z^R$ for all L and R. By definition, $x \prec y$ implies there merely exists R_0 with $x^{R_0} \preceq y$ or L_0 with $x \preceq y^{L_0}$. If $x^{R_0} \preceq y$, then the outer inductive hypothesis yields $x^{R_0} \preceq z$, hence $x \prec z$. If $x \preceq y^{L_0}$, then the inner inductive hypothesis for $y^{L_0} < z$ (which holds by the constructor of $y \leq z$) yields $x \prec z$.

This completes the inner induction. Thus, for any x defined by a cut, we have $(x \prec -)$ and $(x \preceq -)$ defined by (11.6.12) and (11.6.13), and transitive on the right.

To complete the outer recursion, we need to verify these definitions are transitive on the left. After a No-induction on z, we end up with three cases that are essentially identical to those just described above for transitivity on the right. Hence, we omit them. □